

```
- name: Update the software package repository
  yum:
    name: '*'
    update_cache: yes

- name: Install Graphite server
  package:
    name: "{{ item }}"
    state: latest
  with_items:
    - graphite-web
```

We first import the keys for the Extra Packages for Enterprise Linux (EPEL) repository and update the software package list. The 'graphite-web' package is then installed using Yum. The above playbook can be invoked using the following command:

```
$ ansible-playbook -i inventory/kvm/inventory playbooks/
configuration/graphite.yml --tags "graphite"
```

MySQL

A backend database is required by Graphite. By default, the SQLite3 database is used, but we will install and use MySQL as shown below:

```
- name: Install MySQL
  hosts: graphite
  become: yes
  become_method: sudo
  gather_facts: true
  tags: [database]

tasks:
- name: Install database
  package:
    name: "{{ item }}"
    state: latest
  with_items:
    - mysql
    - mysql-server
    - MySQL-python
    - libselinux-python

- name: Start mysqld server
  service:
    name: mysqld
    state: started

- wait_for:
  port: 3306

- name: Create graphite database user
  mysql_user:
```

```
name: graphite
password: graphite123
priv: '*.*:ALL,GRANT'
state: present

- name: Create a database
  mysql_db:
    name: graphite
    state: present

- name: Update database configuration
  blockinfile:
    path: /etc/graphite-web/local_settings.py
    block: |
      DATABASES = {
        'default': {
          'NAME': 'graphite',
          'ENGINE': 'django.db.backends.mysql',
          'USER': 'graphite',
          'PASSWORD': 'graphite123',
        }
      }

- name: syncdb
  shell: /usr/lib/python2.6/site-packages/graphite/
manage.py syncdb --noinput

- name: Allow port 80
  shell: iptables -I INPUT -p tcp --dport 80 -m state
--state NEW,ESTABLISHED -j ACCEPT

- name:
  lineinfile:
    path: /etc/httpd/conf.d/graphite-web.conf
    insertafter: '        # Apache 2.2'
    line: '                Allow from all'

- name: Start httpd server
  service:
    name: httpd
    state: started
```

As a first step, let's install the required MySQL dependency packages and the server itself. We can then start the server and wait for it to listen on port 3306. A graphite user and database is created for use with the Graphite Web application. For this example, the password is provided as plain text. In production, use an encrypted Ansible Vault password.

The database configuration file is then updated to use the MySQL credentials. Since Graphite is a Django application, the *manage.py* script with *syncdb* needs to be executed to create the necessary tables. We then allow port 80 through the firewall in order to view the Graphite dashboard. The *graphite-web.conf* file is updated to allow read access, and the